



COLUMBIA | ENGINEERING

The Fu Foundation School of Engineering and Applied Science

APPLIED PHYSICS AND APPLIED MATHEMATICS
WITH MATERIALS SCIENCE AND ENGINEERING

APPH 6081

Solid State Physics HW 7

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1 Problem I

(a) From last homework, we derive the total Hamiltonian under the orbital basis, written as

$$\begin{aligned} H_k &= \begin{pmatrix} \varepsilon_d & t_{pd} \\ t_{pd} & \varepsilon_p \end{pmatrix} + \exp(ika) \begin{pmatrix} 0 & 0 \\ t_{pd} & 0 \end{pmatrix} + \exp(-ika) \begin{pmatrix} 0 & t_{pd} \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} \varepsilon_p + 1 & t_{pd}(1 + \exp(-ika)) \\ t_{pd}(1 + \exp(ika)) & \varepsilon_p \end{pmatrix}. \end{aligned} \quad (1)$$

So the eigenvalues of the Hamiltonian is

$$\varepsilon(k) = \varepsilon_p + \frac{1}{2} \pm \frac{1}{2} \sqrt{\cos^2 \frac{ka}{2} + 1}. \quad (2)$$

Select $\varepsilon_p = \frac{1}{2}$, then see Fig. 1 for the band structure plot of the result in the first BZ.

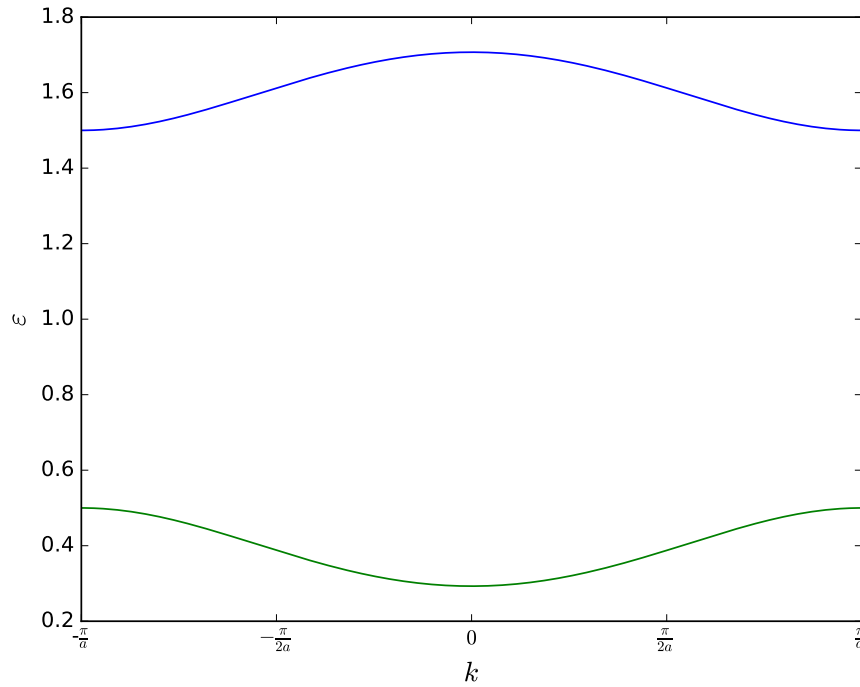


Figure 1: Band structure plot of the result in the first BZ.

(b) Derive $\nabla_k \varepsilon(\mathbf{k})$ we have

$$\begin{aligned}\nabla_k \varepsilon(\mathbf{k}) &= \pm \frac{2at^2 \sin(ak)}{\sqrt{8t^2(\cos(ak) + 1) + 1}} \\ &= \pm \frac{a\sqrt{(4(\varepsilon - \varepsilon_p + \frac{1}{2})^2 - 1)(-4(\varepsilon - \varepsilon_p + \frac{1}{2})^2 + 16t^2 + 1)}}{8(\varepsilon - \varepsilon_p + \frac{1}{2})},\end{aligned}\quad (3)$$

if we let $\varepsilon_p - \frac{1}{2} = 0$, then

$$\nabla_k \varepsilon(\mathbf{k}) = \pm \frac{a\sqrt{(4\varepsilon^2 - 1)(-4\varepsilon^2 + 16t^2 + 1)}}{8\varepsilon}.\quad (4)$$

so the DOS is

$$\begin{aligned}g(\varepsilon) &= \frac{1}{\pi} \int_{S_n(\varepsilon)} dS \frac{1}{\left| \frac{a\sqrt{(4\varepsilon^2 - 1)(-4\varepsilon^2 + 16t^2 + 1)}}{8\varepsilon} \right|} \\ &= \frac{1}{\pi} \int_{S_n(\varepsilon)} dS \left| \frac{8\varepsilon}{a\sqrt{(4\varepsilon^2 - 1)(-4\varepsilon^2 + 16t^2 + 1)}} \right| \\ &= \frac{16\varepsilon}{a\sqrt{(4\varepsilon^2 - 1)(-4\varepsilon^2 + 16t^2 + 1)}}\end{aligned}\quad (5)$$

Or we have

$$N(\varepsilon) = \frac{2}{V_{\text{FBZ}}} \int dk \theta(\varepsilon - \varepsilon(k)),\quad (6)$$

where $V_{\text{FBZ}} = \frac{2\pi}{a}$, $\varepsilon(k)$ is given by (2). So

$$\begin{aligned}N(\varepsilon) &= \frac{2}{\frac{2\pi}{a}} \int dk \theta\left(\varepsilon - \varepsilon_p - \frac{1}{2} \pm \frac{1}{2} \sqrt{\cos^2 \frac{ka}{2} + 1}\right) \\ &= \frac{a}{\pi} k \Big|_{-\frac{1}{a} \arccos(8(\varepsilon-1)^2-3)}^{\frac{1}{a} \arccos(8(\varepsilon-1)^2-3)} \\ &= \frac{2}{\pi} \arccos(8(\varepsilon - 1)^2 - 3).\end{aligned}\quad (7)$$

So the DOS is the derivative of $N(\varepsilon)$, written as

$$\begin{aligned}g(\varepsilon) &= \frac{dN}{d\varepsilon} \\ &= -\frac{32(\varepsilon - 1)}{\pi \sqrt{1 - (8(\varepsilon - 1)^2 - 3)^2}}.\end{aligned}\quad (8)$$

Here ε ranges from 0.292893 to 0.5, for ε ranges from 1.5 to 1.70711, the density of states is

$$g(\varepsilon) = \frac{dN}{d\varepsilon} = \frac{32(\varepsilon - 1)}{\pi\sqrt{1 - (8(\varepsilon - 1)^2 - 3)^2}}. \quad (9)$$

So plot as Fig. 2.

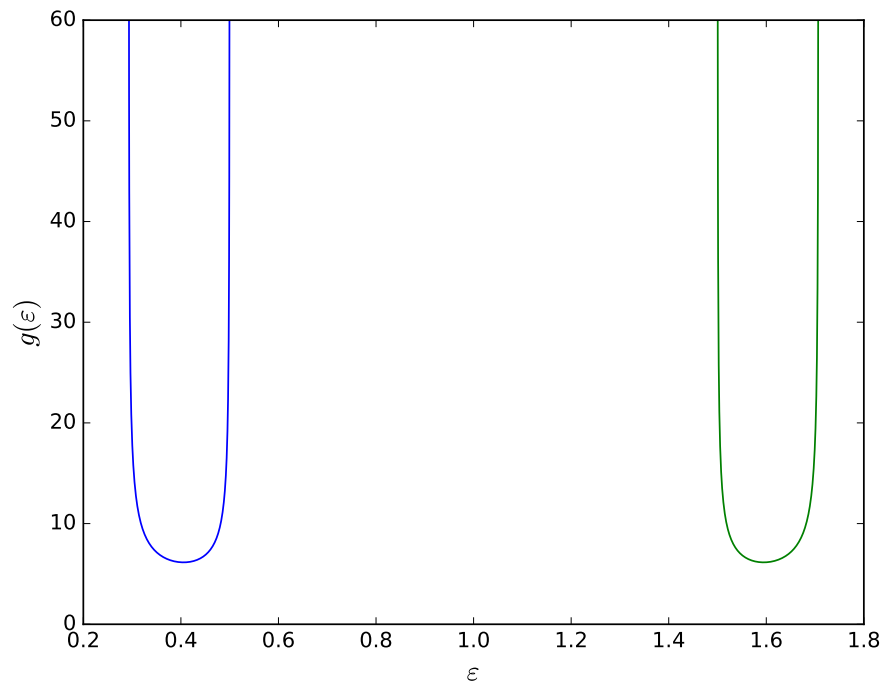


Figure 2: DOS.

(c) Suppose $|\psi_k\rangle = c_1 |d\rangle + c_2 |p\rangle$. So the eigenvector satisfies the equation

$$(H_k - \varepsilon I) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0, \quad (10)$$

solve this equation, we will derive c_1 and c_2 , so

$$\begin{aligned} \langle d|\psi\rangle &= c_1, \\ \langle p|\psi\rangle &= c_2. \end{aligned} \quad (11)$$

So we have

$$\begin{aligned} g_d(\varepsilon) &= g(\varepsilon)c_1^*c_1, \\ g_p(\varepsilon) &= g(\varepsilon)c_2^*c_2. \end{aligned} \quad (12)$$

See Fig. 3 for reference. Notice here d-state DOS plus p-state DOS equal to original DOS, to make it convenient to observe, I add 1 to original DOS to distinguish d-state DOS plus p-state DOS.

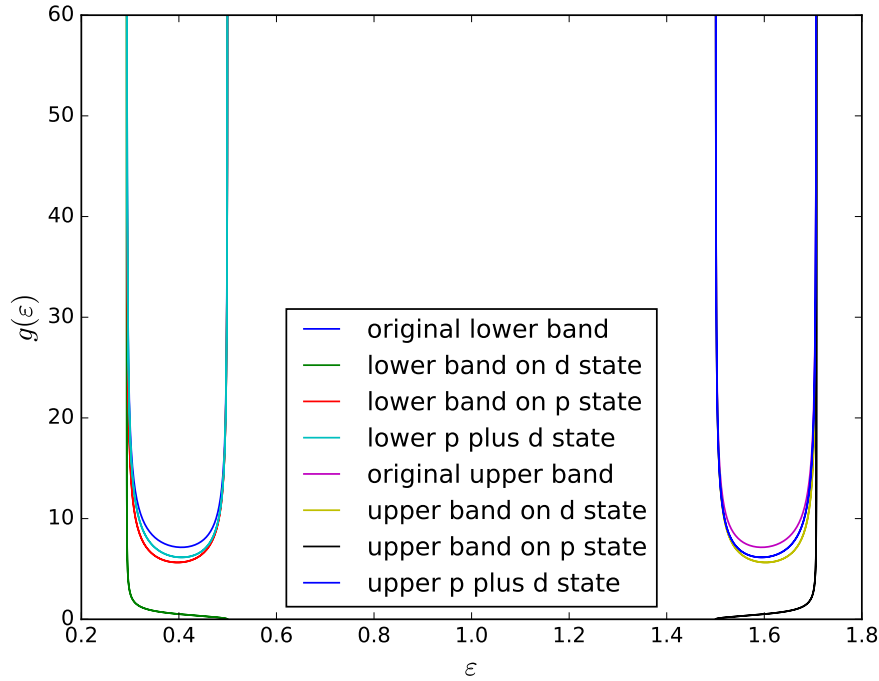


Figure 3: Projected DOS as well as original DOS.

(d) Assuming $\varepsilon_p = \frac{1}{2}$, Expand eigenvalue by assuming t_{pd} is a very small number. The analytical eigenvalues are

$$\begin{aligned} \varepsilon_1 &= \frac{1}{2} + \frac{1}{2} + \frac{1}{2}\sqrt{8t^2(\cos(ka) + 1) + 1}, \\ \varepsilon_2 &= \frac{1}{2} + \frac{1}{2} - \frac{1}{2}\sqrt{8t^2(\cos(ka) + 1) + 1}. \end{aligned} \quad (13)$$

So by Taylor expansion, we derive

$$\begin{aligned} \varepsilon_1 &\approx 2t^2(\cos(ka) + 1) + \frac{3}{2}, \\ \varepsilon_2 &\approx -2t^2(\cos(k) + 1) + \frac{1}{2}. \end{aligned} \quad (14)$$

Then plot them as Fig. 4.

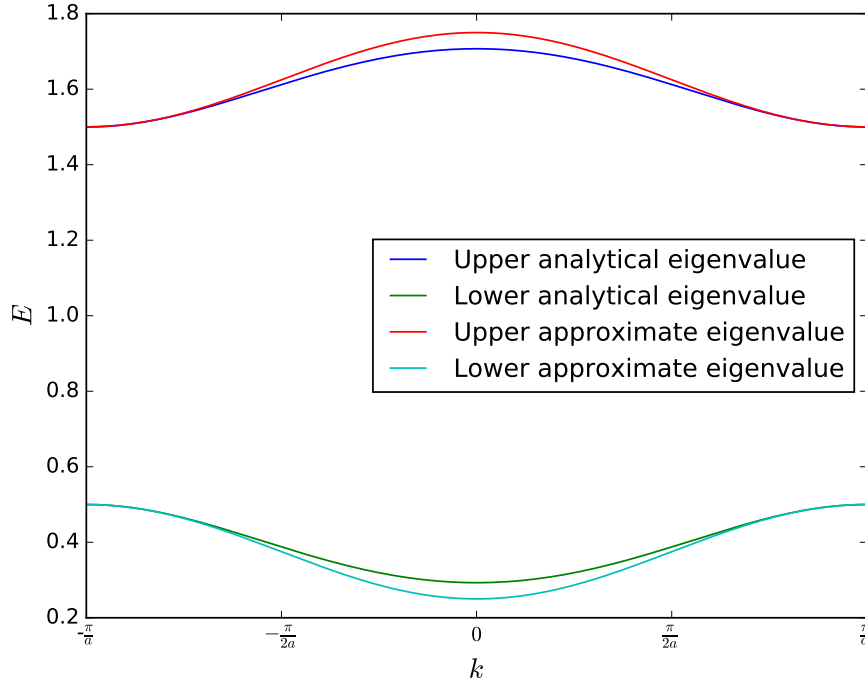


Figure 4: Analytical eigenvalue and approximate eigenvalue.

- (e) In the lower band, 2 electrons were filled, and in upper bands, there fills 1 electron. So ε_F satisfy following equation

$$N(\varepsilon_F) = \frac{2}{\pi} \arccos(8(\varepsilon_F - 1)^2 - 3) = 1, \quad (15)$$

solve the equation, we derive $\varepsilon_F = 1.61237 \text{ eV}$.

2 Problem II

- (a) Draw the graphene structure as Fig. 5, with dashed line denoting a unit cell of graphene. Here the primitive vectors are $\mathbf{a}_1$ and $\mathbf{a}_2$, the primitive reciprocal vectors are $\mathbf{b}_1$ and $\mathbf{b}_2$, a is the length of a

hexagonal edge. If we set origin at the point where writes Γ , and use Cartesian coordinates, then

$$\begin{cases} \mathbf{a}_1 = (\frac{3a}{2}, -\frac{\sqrt{3}a}{2}), \\ \mathbf{a}_2 = (\frac{3a}{2}, \frac{\sqrt{3}a}{2}), \end{cases} \quad (16)$$

$$\begin{cases} \mathbf{b}_1 = (\frac{2\pi}{3a}, -\frac{2\pi}{\sqrt{3}a}), \\ \mathbf{b}_2 = (\frac{2\pi}{3a}, \frac{2\pi}{\sqrt{3}a}). \end{cases} \quad (17)$$

So in each cell, there are 2 atomic orbitals, the red orbital and the green orbital. For green orbital,

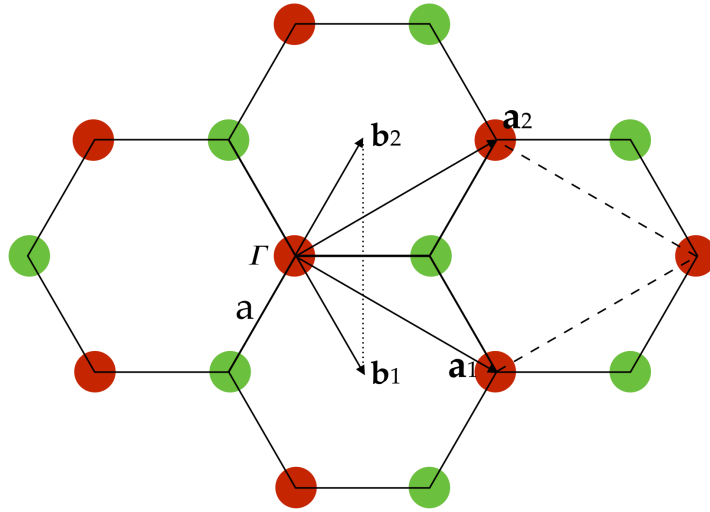


Figure 5: The honeycomb structure of graphene.

consider the eigenfunction of translation operator to be

$$\begin{aligned} \psi(\mathbf{k}, \mathbf{r}) &= \frac{1}{\sqrt{1 \times 3}} \sum_v \sum_{\mathbf{R}} \exp(i\mathbf{k} \cdot \mathbf{R}) \phi_v(\mathbf{r} - \mathbf{R}) \\ &= \frac{1}{\sqrt{3}} \left(\phi_1(\mathbf{r}) + \sum_{\mathbf{R}} \exp(i\mathbf{k} \cdot \mathbf{R}) \phi_2(\mathbf{r} - \mathbf{R}) \right), \end{aligned} \quad (18)$$

where $\phi_1(\mathbf{r})$ denotes green orbital itself, $\phi_2(\mathbf{r})$ denotes 3 nearest red orbitals. Similarly, for red orbital, the eigenfunction of translation operator is

$$\psi(\mathbf{k}', \mathbf{r}) = \frac{1}{\sqrt{3}} \left(\phi_2(\mathbf{r}) + \sum_{\mathbf{R}'} \exp(i\mathbf{k}' \cdot \mathbf{R}') \phi_1(\mathbf{r} - \mathbf{R}') \right). \quad (19)$$

So the Hamiltonian is

$$H_k = \sum_{\mu, \nu} \sum_{\mathbf{R}} \exp(i\mathbf{k} \cdot \mathbf{R}) \langle \phi_{\mu}(\mathbf{r}) | H | \phi_{\nu}(\mathbf{r} - \mathbf{R}) \rangle, \quad (20)$$

where

$$\begin{aligned}
 \langle \phi_1(\mathbf{r}) | H | \phi_1(\mathbf{r}) \rangle &= 0, \\
 \langle \phi_1(\mathbf{r}) | H | \phi_2(\mathbf{r} - (-a, 0)) \rangle &= \langle \phi_1(\mathbf{r}) | H | \phi_2(\mathbf{r} - (\frac{1}{2}a, -\frac{\sqrt{3}}{2}a)) \rangle = \langle \phi_1(\mathbf{r}) | H | \phi_2(\mathbf{r} - (\frac{1}{2}a, \frac{\sqrt{3}}{2}a)) \rangle = t, \\
 \langle \phi_2(\mathbf{r}) | H | \phi_1(\mathbf{r} - (a, 0)) \rangle &= \langle \phi_2(\mathbf{r}) | H | \phi_1(\mathbf{r} - (-\frac{1}{2}a, \frac{\sqrt{3}}{2}a)) \rangle = \langle \phi_2(\mathbf{r}) | H | \phi_1(\mathbf{r} - (-\frac{1}{2}a, -\frac{\sqrt{3}}{2}a)) \rangle = t, \\
 \langle \phi_2(\mathbf{r}) | H | \phi_2(\mathbf{r}) \rangle &= 0.
 \end{aligned} \tag{21}$$

Written as matrix form, it will be

$$H_{\mathbf{k}} = \begin{pmatrix} 0 & h \\ h^* & 0 \end{pmatrix}, \tag{22}$$

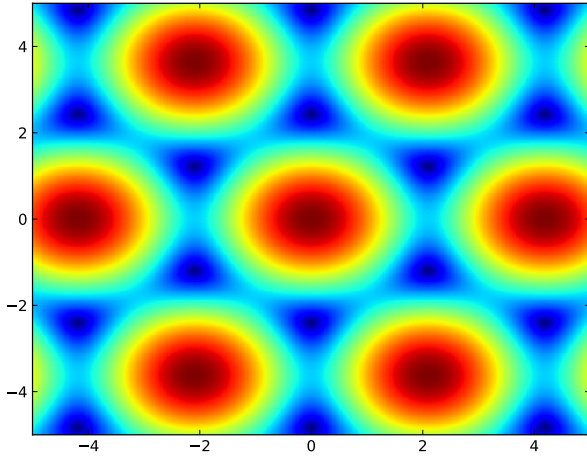
where $h = t \left(e^{-ik_x a} + e^{i\left(\frac{1}{2}k_x a - \frac{\sqrt{3}}{2}k_y a\right)} + e^{i\left(\frac{1}{2}k_x a + \frac{\sqrt{3}}{2}k_y a\right)} \right)$, and h^* is its complex conjugate. So the analytical eigenvalues are

$$\begin{aligned}
 E_{\pm} &= \pm \sqrt{t^2 h^* h} \\
 &= \pm 2.5 \sqrt{4 \cos\left(\frac{3k_x a}{2}\right) \cos\left(\frac{\sqrt{3}k_y a}{2}\right) + 2 \cos\left(\sqrt{3}k_y a\right) + 3}
 \end{aligned} \tag{23}$$

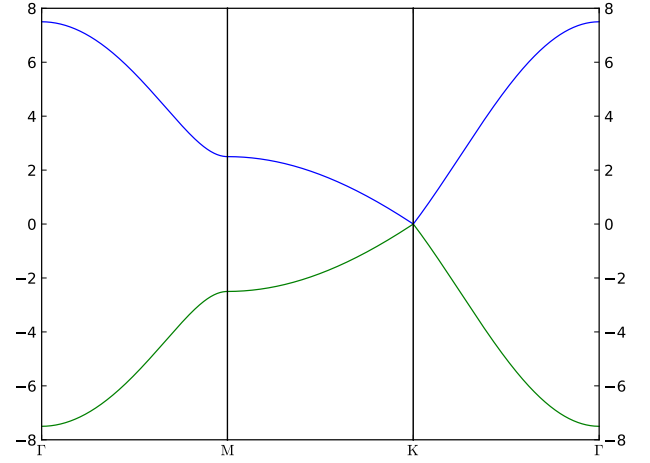
- (b) The coordinates (k_x, k_y) of Γ , M and K points are $(0, 0)$, $(\frac{2\pi}{3a}, 0)$, and $(\frac{2\pi}{3a}, \frac{2\pi}{3\sqrt{3}a})$, respectively. See Fig. 6(b) for energy along the k-path.
- (c) Because there are 2 electrons per unit cell, assuming N unit cells in the crystal, so there are $2N$ electrons in the crystal. Those electrons fully filled the lowest band in first BZ. So the Fermi energy is the maximum energy of the lowest band. From the figure we can see it is at K point, so the Fermi energy is 0 eV.
- (d) Let $k'_x = \frac{2\pi}{3a} + dk_x$, $k'_y = \frac{2\pi}{3\sqrt{3}a} + dk_y$, substitute into (23), and use approximation $\cos(x) \approx 1 - \frac{x^2}{2}$ and $\sqrt{1+x} \approx 1 + \frac{1}{2}x$, we derive

$$\begin{aligned}
 \varepsilon &= \pm 2.5 \sqrt{4 \cos\left(\frac{3(\frac{2\pi}{3a} + dk_x)a}{2}\right) \cos\left(\frac{\sqrt{3}(\frac{2\pi}{3\sqrt{3}a} + dk_y)a}{2}\right) + 2 \cos\left(\sqrt{3}\left(\frac{2\pi}{3\sqrt{3}a} + dk_y\right)a\right) + 3} \\
 &\approx \pm \frac{3ta}{2} \sqrt{dk_x^2 + dk_y^2} \\
 &= \pm \frac{3ta}{2} k.
 \end{aligned} \tag{24}$$

So the dispersion relation is linear near the K point.



(a) Density plot of graphene band structure.



(b) Energy along k-path.

Figure 6: Some plots of graphene band structure.

(e) We have

$$N(\varepsilon) = \frac{2}{V_{\text{FBZ}}} \int d\mathbf{k} \theta(\varepsilon - \varepsilon(\mathbf{k})), \quad (25)$$

where $V_{\text{FBZ}} = \frac{8\pi^2}{3\sqrt{3}a^2}$ is the volume of the first BZ, and θ is Heaviside step function. For a certain ε , only $\varepsilon(\mathbf{k}) < \varepsilon$ can be counted. Because in this question, the electrons fully filled the first band, and in the first band there are 2 electrons. Thus, if we want to count the electrons within energy interval $\varepsilon < \varepsilon(\mathbf{k}) < \varepsilon_F$, we use

$$\begin{aligned} N'(\varepsilon) &= \frac{2}{V_{\text{FBZ}}} \int \theta(\varepsilon - \varepsilon(\mathbf{k})) d\mathbf{k} \\ &= \frac{2}{V_{\text{FBZ}}} \int_{|\varepsilon(\mathbf{k})| < |\varepsilon|} d\mathbf{k} \\ &= 2 - \frac{2}{V_{\text{FBZ}}} \int_{|\varepsilon| \leq |\varepsilon(\mathbf{k})| \leq |\varepsilon_F|} d\mathbf{k} \\ &= 2 - \frac{2}{V_{\text{FBZ}}} \int_{0 < k_x^2 + k_y^2 < k^2} dk_x dk_y \\ &= 2 - \frac{2}{V_{\text{FBZ}}} \pi k^2 \\ &= 2 - \frac{3\sqrt{3}a^2 k^2}{4\pi}. \end{aligned} \quad (26)$$

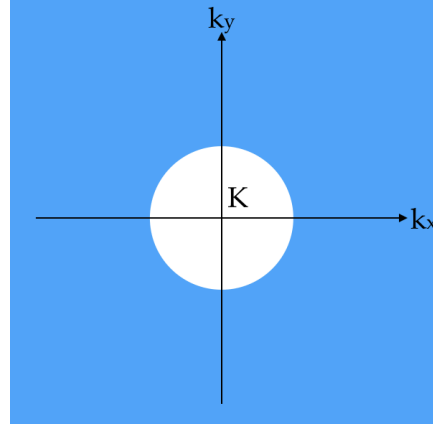


Figure 7: Schematic plot of filled band with K point as origin.

See Fig. 7, within the white circle $k_x^2 + k_y^2 < k^2$, those are the unfilled region whose energy is above ε . However, near the K point, as above,

$$k = \frac{2\varepsilon}{3at}, \quad (27)$$

so

$$N'(\varepsilon) = 2 - \frac{\varepsilon^2}{\sqrt{3}\pi t^2}, \quad (28)$$

So the DOS is the derivative of $N'(\varepsilon)$, written as

$$\begin{aligned} g(\varepsilon) &= \frac{dN'(\varepsilon)}{d\varepsilon} \\ &= -\frac{2\varepsilon}{\sqrt{3}\pi t^2}. \end{aligned} \quad (29)$$

Here ε is below ε_F (0 eV), so $g(\varepsilon)$ is still greater than 0.